

DO REPS TO **FLY.**

FitFlap makes your **body the controller**. Strap a phone to your arm, raise and lower it, and pilot a bird through obstacles in real time — the workout **is** the game, not a tracker you check afterward. Our hackathon demo, **Flappy Workout**, runs entirely in the browser with an iPhone as the controller and a laptop as the screen.

THE PROBLEM

People quit workouts because they feel **repetitive and lonely**. Fitness apps track **after** you sweat — they don't make the set itself fun. VR fitness does, but it's expensive, bulky, and overkill for a quick daily burn.

HOW IT WORKS

- 1 Open on a laptop.** The game shows a QR code — no app, no install.
- 2 Scan with your iPhone.** It pairs over a WebSocket relay and becomes your motion controller.
- 3 Calibrate in 2 taps** (arm low / high). Now your arm drives the bird's height live.
- 4 One up-and-down curl = 1 rep.** Thread the gaps, bank a score, climb the leaderboard.

WHY IT'S DIFFERENT

- ▶ **The workout is the game** — engagement happens during the set, not in a post-hoc dashboard.
- ▶ **No headset, no console, no install** — just the phone already on your arm and any screen.
- ▶ **A platform, not a game** — one motion controller spanning a library of workouts. Flappy Workout is Game 01.

♦ WHAT WE SHIPPED — LIVE DEMO

- ✓ **iPhone ↔ laptop**, paired by QR over a WebSocket relay — 100% browser, zero install.
- ✓ **Drift-free, mount-agnostic motion** — arm height from a gravity-vector sweep; works however the phone is strapped.
- ✓ **Real rep detection** via hysteresis on a full up→down cycle.
- ✓ **Difficulty that scales the workout** — gaps zig-zag and speed ramps, forcing wider arm sweeps.
- ✓ **Volume tracking** — reps × weight = total lb / kg lifted.
- ✓ **Live shared leaderboard** + a mouse fallback for instant demos.
- ✓ **One-command deploy** to Google Cloud Run (auto-HTTPS, which is what unlocks iOS motion).

♦ AI · GEMINI

A tap calls **gemin-2.5-flash** to mint a fresh challenge — a catchy **title**, a hype **line**, and a **rep goal** (10–50) — different every round. A built-in fallback means the demo **never breaks**, even offline.

Steady Wings — 35 smooth raises in 90s. Control the shake, not just the rep.

~60Hz

MOTION STREAMED

2

TAPS TO CALIBRATE

0

INSTALLS NEEDED

1

PHONE = CONTROLLER

∞

GAMES ON PLATFORM

MARKET

Fitness apps × wearables × mobile gaming — and **everyone already owns the sensor** in their pocket.

BUSINESS MODEL

Free app → **premium modes**, **AI plans**, **friend leagues**, optional wearable, and school / gym / corporate deals.

GO-TO-MARKET

Inherently shareable — people sweating & laughing on camera → **Reels / TikTok / Shorts**, friend challenges, dorm & gym micro-leagues.

VISION

The easiest way to turn **any short workout into a game**. Open it to play; finish realizing you just trained.

You came to play. You left stronger.

NODE · EXPRESS · WEBSOCKET · CANVAS · GEMINI · CLOUD RUN